

Classifying Music Genre and Recommending songs using various ML Algorithms

Devapatla Bhargavi, Chandini Magoo, Manpreet Kaur

bhargavir139@gmail.com, chandnimagoo@mru.edu.in, manpreet@mru.edu.in

Department of Computer science & Engineering

Manav Rachna University

Faridabad, India

Abstract—The task of music genre classification in the realm of Music Information Retrieval is considered very challenging because audio features have varied complexity and overlapping nature. This paper addresses the issue by employing, comparing, and experimenting on diverse machine learning and deep learning models, including XGBoost, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Convolutional Neural Networks (CNN). The experiments are conducted on the GTZAN dataset, which has widely been used as a benchmark for music genre classification. Among the models implemented, the XGBoost demonstrated the best performance in predictions of the genre labels, acquiring a higher degree of accuracy as compared to the classical machine learning techniques. Besides, the study includes the development of a music recommendation system that suggests songs based on the input audio. It indicates that machine learning methods, like XGBoost, are highly effective for genre classification, but further enhancements, such as feature optimization and dataset expansion, could improve the performance of more traditional models.

Keywords—*Music Information Retrieval, Music Genre Classification, Convolutional Neural Networks, XGBoost, Random Forest, K-Nearest Neighbors, Support Vector Machines.*

1. INTRODUCTION

Classification of music genres is a critical task with several real-world applications. Because the number of music released each day increases relentlessly, particularly on websites like SoundCloud and Spotify, the need for proper metadata for database management, searching, and storage will also grow more. Music streaming and selling businesses require efficient genre identification to organize playlists and collections, while accurate music labeling offers limitless possibilities for statistical analysis and user insights. Music genre classification involves two essential stages: feature extraction and classification. For feature extraction, CSV file is used containing audio features extracted from the dataset. For classification, we relied on two approaches: basic classification algorithms and a deep learning strategy. Audio features are inputs to four machine learning models: K-Nearest Neighbors, Random Forest, Support Vector Machines, and XGBoost in the classification approach, which directly predict music genre from the features provided. In the deep learning approach, CNN is classified on audio features produced by the mel spectrogram representation of the audio files. The output of the model is the predicted music genre. This study shows the effectiveness of machine learning and sophisticated algorithms, such as XGBoost, in attaining coherent music genre classification.

2. LITERATURE SURVEY

The researchers in [1] used supervised machine learning techniques like the Gaussian Mixture Model and k-nearest neighbour classifiers to solve this problem. Timbral structure, rhythmic content, and pitch content are the three features introduced. On ten genres, 61% accuracy was achieved. As part of the MARSYAS framework, a free software platform for developing and evaluating computer audio applications, the entire feature set is now available to the general audience. To study the difficulty of feature selection in the AMGC problem, [2] use a large number of classifiers. They use linear and quadratic discriminant classifiers, as well as the Nave-Bayes classifier and k-NN variants. For feature extraction, they use the GTZAN database and the MARSYAS framework. For each of the classifiers, the features were examined according to groups. In order to discover the optimal feature set for the problem, they use the Forward Feature Selection (FFS) and Backward Feature Selection (BFS) approaches. They report that using the feature selection approach, they were able to achieve positive classification findings. Hareesh Bahuleyan used various machine learning techniques to classify the music genre [3]. In this study, some of the traditional machine learning techniques and CNN were used in order to complete the task. For their model, the ensemble of CNN and XGBoost proved to be beneficial. The research in [4] focuses on the relative relevance of the traits. They use features from three-time scales: (a) short-term, (b) middle-term, and (c) long-term. They utilize a single layer neural net and a Gaussian classifier based on the covariance matrix as their classifiers. Authors ran tests on two datasets, the first of which contained 100 music pieces from five different genres and the second of which contained 354 music pieces from six different genres. The findings of the trials demonstrate that combining intermediate and long-term features improves classification accuracy. The literature has been reviewed in music genre classification where several machine learning and deep learning techniques have been proved to be capable of enhancing the classification accuracy. One study applied a VGG-16 CNN model to attain excellent accuracy but did not specify the accuracy value [5]. The other study reported that the accuracy of the classifier was 84% at the beginning but improved after further refinement. The second work was based on the classification of Indian Hindi songs into Sufi and Classical genres, with a reported accuracy of around 85%. Traditional algorithms were utilized, such as K-NN, Decision Trees, and SVM, in addition to CNNs for feature extraction purposes. Performance measures such as

accuracy, precision, recall, and F1 score were used to evaluate the models. The results suggest that deep models, in particular the CNNs, are promising but require much improvement as for robustness and generalization capabilities across datasets. Music genre classification and recommendation are of paramount importance in the digital music environment. This study used SVM and K-NN for genre classification from the GTZAN dataset [6]. It used a technique in the SVM algorithm that resulted in the highest successful classification. Traditional methods were outperformed. The high-level features and sequential dependencies in audio data were also captured with deep learning techniques, such as CNN and Long Short-Term Memory (LSTM) networks. The LSTM model achieved a classification accuracy of 89.71% on average. Accuracy, precision, recall, and F1 score are amongst the performance metrics used to report an assessment of the models' efficiency. It can be concluded that incorporating diverse machine learning techniques can be enhanced for genre classification. Possible future improvements in this model may involve real-time genre recognition as well as generative models for music transformation. The project also found potential for transfer learning applications in the areas of improving classification accuracy across different datasets. The classification of the genres of music plays a vital role in enhancing user experience in music streaming services. For this study, a CNN model [7] was used on audio signals from the GTZAN dataset compared to other traditional machine learning algorithms such as K-NN and SVM. It included audio samples transformed into spectrogram images, which were fed to the CNN to help effectively learn high-level features from it. All performance measures including accuracy, precision, recall, and confusion matrix were used to evaluate all models' reliability. In comparison with these traditional methods, CNN had a far better result, where accuracy improved significantly with increased dataset size. Note that the accuracy rose from 83% to 91% with additional training data. This emphasizes the need for volume in datasets while training deep learning models for music genre classification. Further work would be to apply CNN on other datasets and test it on real-time music recommendation systems and affirm its efficacy in this application. The two applications, classification and recommendation, of music genres are essential due to the ever-increasing volume of music available online. This paper employed CNNs in the classification of music songs with audio features with algorithms like K-NN, Decision Tree, Logistic Regression, and SVM[8]. Training was undertaken with the GTZAN dataset, holding about 1,000 songs sampled for 30 seconds. The performance parameters here used include accuracy, precision, recall, F1 score, and AUC-ROC curve to check the model's performance. The CNN method returned a high accuracy ratio as compared to the conventional methods. This research highlights the potential for real-time music recommendation systems, enhancing user experience through personalized song suggestions. Future work will focus on expanding the dataset and improving model performance for dynamic applications.

The classification and recommendation of music genres are vital in the digital music era, where users face an abundance

of choices. Using the digital signal processing to acquire the acoustic features, this research used Convolutional Neural Networks (CNN) to classify songs. A training set was taken by using GTZAN dataset close to 1,000 tracks [9]. The performance metrics of the model included accuracy, precision, and recall. The CNN model showed good classification accuracy and was more superior compared to traditional methods such as SVM and K-nearest neighbors (K-NN). This research highlights the potential for real-time music recommendation systems, enhancing user experiences through personalized suggestions. Future work will focus on improving model performance and expanding the dataset. Recent work in [10] focuses on the application of HMMs and acoustic segment modeling in music genre classification. By segmenting musical pieces into acoustic units, akin to phonetic units in language, the study enables a deeper analysis of musical structure. Using a timbral dictionary, HMMs effectively model transitions between musical notes and chords, enhancing classification accuracy. The research utilized a dataset from Magnatunes, dividing songs into 30-second segments. This approach achieved high accuracy rates, with genre classification exceeding 90%. The combination of support vector machines with acoustic segment models was very efficient. This approach shows a promise for applying syntactic rules motivated by "musical language" to improve genre categorization. This study aims at music genre classification and recommendation using machine learning methodologies. Acoustic features such as zero-crossing rate, spectral centroid, contrast, bandwidth, roll off, and MFCC were extracted from raw music signals using digital signal processing[11]. A number of algorithms, including SVM and KNN and Random Forest and Naive Bayes, were experimented with the highest found accuracy using SVM on the GTZAN data set at a rate of 89%. Using CNNs for classifying spectrograms for deep feature extraction resulted in an accuracy of 92%. The experiment also proved that hand-crafted feature concatenation with CNN-based features improved the classification accuracy to 94%. The results stress the power of machine learning and, especially, deep learning in the development of advanced audio classification and music recommendation systems. Random Forests can be viewed as a kernel method for two-class problems, stressing roles of randomness and correlation as means for lifting performance[12]. The framework for Stochastic Discrimination clarifies further what the roles of correlations might be between variables and classifier strength. Empirical studies showed that the mean squared generalization error depends on the correlation of residuals and on the individual performances of the trees. Often, the optimal accuracy is attained by choosing a few features at each node. Random Forests are also resistant to noise and outliers thus competing with algorithms like Adaboost. They also provide internal estimates of error and variable importance. Genre classification and recommendation are critical in the digital music environment. In this research, Support Vector Machines (SVM) and K-Nearest Neighbors (K-NN) were used for genre classification from the GTZAN dataset. The SVM algorithm reached the most accurate levels of classification, a situation that overtakes traditional techniques. CNNs and LSTM are among the deep learning techniques used to extract high-level features from the audio

data. In [13] the mean recognition rate of the LSTM model is 89.71%. There are other indicators to measure the performance of an assessment system, including for example accuracy ratio, recall rate and F1 score. Evaluation metrics used for assessing the efficiency of the model were accuracy, precision, recall, and F1 score. Potential future developments could be in real-time genre identification and the application of transfer learning for improved classification accuracy across different datasets. The study on music genre classification made use of the Free Music Archive small (fma_small) dataset [14] to compare among different models, such as Logistic Regression, Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN). The simple ANN model achieved a test accuracy of 64% while the CNN model excelled at the accuracy of 88.5%, thus emerging to be the topmost among all the spectrogram-based models. Classification based on images was found to outdo classification based on features. In the near future, the datasets can be increased for further boosting the performance of CRNN and CNN-RNN models. The overall conclusion is that deep learning techniques are very effective for music genre classification. This paper focuses on Convolutional Neural Networks [15] to handle the huge databases of digital music. It uses the GTZAN dataset, hitting 76% accuracy rate in genre classification and propagating suitable music with its recommendation system. The conclusion is that CNNs significantly augment music retrieval systems. Future work is developing a GUI and the investigation of hybrid models that might increase the rate of accuracy. A couple of limitations were encountered, including reliance on a specific dataset, potential inaccuracies because of a single model approach, and the requirement of substantial computational resources. Overall, the research demonstrates the power of deep learning in music information retrieval. This work proposed developing a music recommendation system using deep learning techniques with a focus on genre classification.[16] It uses three models: the baseline CNN, a CNN-LSTM for sequence handling, and a CNN-BiLSTM for improved temporal processing. The classification accuracies are 69% for CNN, 72% for CNN-LSTM, and 71% for CNN-BiLSTM with the former being the highest. A content-based recommendation system based on the cosine similarity dealt adequately with the cold-start problem, although the accuracy of instrumental genres was lower because of imbalanced data. The study concludes that deep learning can significantly improve music recommendations and suggests future work on hybrid systems incorporating user data. Limitations include the lack of user interaction data and lower accuracy for certain genres. Overall, the research highlights the potential of automated systems in enhancing user satisfaction in the music industry. Music genre classification and recommendation systems use machine learning and deep learning technologies such as SVM, CNNs and LSTMs to achieve high accuracy. CNN-LSTM models are results in 72% accuracy. CNN models referred to as Spectrogram models have good performance, reaching as high as 88.5% accuracy. Cosine similarity is implemented for cold-start problems in content-based systems but not for imbalanced instrumental genre data. The accuracy, precision, recall and F1 measures of models are used to evaluate them. Future work focuses on hybrid systems,

expanding datasets and real time genre identification. The aim of the paper is to propose a new method for the classification of music genres based on sparse representations of auditory temporal modulations [17], and the results were 91% accurate on the GTZAN dataset and 93.56 % accurate on ISMIR2004 Genre dataset. For this purpose, the authors use the auditory temporal modulation representation and sparse representation-based classifiers, as well as some other ways of dimensionality reduction such as NMF and PCA to extract meaningful features. Its constraints are the high requirement for the amount of training data and the diversity of the data samples across the styles. Future research directions include enhancement of the robustness of the models, more hybrid approaches, and additional features for the better classification performance. The paper, on automatic music genre classification, situates that organizational value lies in the management of digital music [18]. Use of Machine Learning Algorithms: Features from frequency, temporal, and cepstral domains. The best classification results were attained for GTZAN at 78% and for ISMIR2004 at 81%. Ensemble methods further boosted performance by three percent on average. The limitations include dependence upon certain datasets and the meager enhancement of certain classifiers. A potential future work would be the real-time genre recognition and transfer learning that can enhance accuracy over numerous datasets. Better feature extraction may be achieved using advanced deep learning methods like CNNs and LSTMs. This paper introduces a new approach of music genre classification using an enhanced Ridgelet Neural Network coupled with an improved IPREO optimizer[19], converting the audio signals into sound spectra for feature extraction. The RNN successfully learns complex features from the sound spectrum, showing noise and variants in audio signal resistance. Experimental results on the GTZAN dataset demonstrated an accuracy of 96% for the proposed model, thereby outperforming traditional approaches and other architectures deep learning. Other performance metrics such as precision, recall, and the F1 score were reported to be 0.96. The limitation of this approach is that it's highly dependent on specific datasets, which risks generalization. An extension of this work will consider how different subgenres have an influence on feature extraction, which might augment the ability of these models to adapt. The system of music genre classification developed with techniques based on deep learning will use the Keras framework. A model will be trained using a dataset split into 60 percent for training, 20 percent for testing, and 20 percent for validation. The proposed model achieves a classification accuracy of 71.66% and, hence, will predict 60 percent of new audio files accurately [20]. Limitations include the lack of user interaction data and lower accuracy for certain genres. Future work is suggested to explore different machine learning algorithms and increase the dataset size. The findings emphasize the potential of automated systems in enhancing music genre classification. Overall, the study contributes valuable insights into the application of deep learning in the music industry.

3. METHODOLOGY

The Dataset used is the GTZAN dataset from Kaggle. The GTZAN dataset is the public dataset that is most frequently used for music genre recognition assessment in machine learning research (MGR). To present a variety of records, files were recorded from 2000 to 2001 from a series of sources, including personal CDs, radios and microphones.

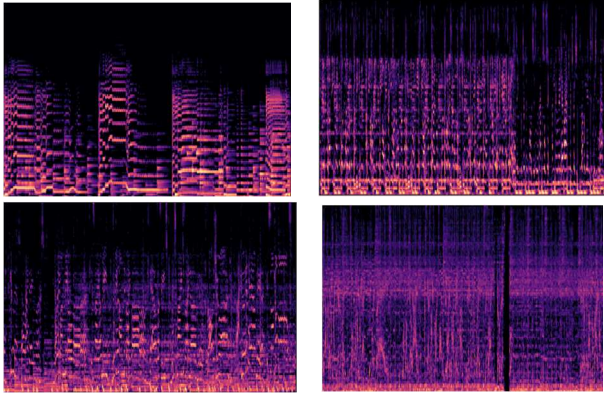


Fig 1: Images of genre from dataset

It contains 4 files in total out of which 2 are extracted features of the audio files, one is the set of 1000 30-second audio clips, all labeled as one out of 10 possible genres and presented as .au files and the last one is the Mel Spectrogram representation of the music files. Folders - The first folder has ten genres, each with 100 audio files, all of which are 30 seconds long, and the second folder contains the visual representation for each audio file; the audio files were converted to Mel Spectrograms for this.

CSV files - Each song (30 seconds long) has a mean and variance computed across several features taken from an audio file in one file. The other file has the same format as the first, except the songs were previously separated into three-second audio files (this way increasing 10 times the amount of data we feed into our classification models).

The 10 different genres in which our audio files are divided are jazz, pop, reggae, country, classical, hip-hop, metal, rock, blues and disco. The size of the image here is a glimpse of the Mel spectrogram representation of the audio file. Some of the features that were used by the model are spectral centroid, spectral bandwidth and Mel spectrogram representation. The dataset is divided in the ratio 8: 2 for train and test. For preprocessing, since our dataset contains images and audio files, it was not possible to check for NULL values or duplication. Principal Component Analysis is used to reduce the dimension of our dataset to 2 features. PCA is an unsupervised learning approach used in machine learning to reduce dimensionality. It's a statistical procedure for converting correlated feature observations into a set of linearly uncorrelated features. In order to do this, we first scale our data using a minmax scaler. Such features are then categorized using a number of algorithms in machine learning into predetermined categories that are based on the

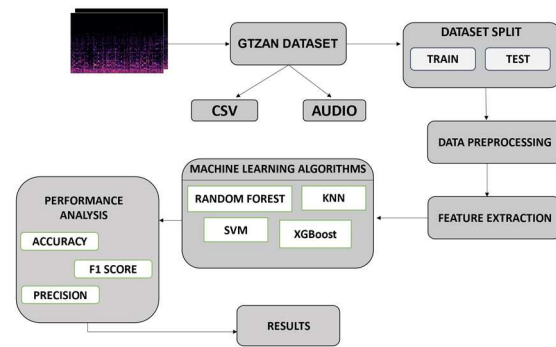


Fig 2: Methodology for Classifying music genre and recommending songs.

processed audio data: Random Forest, K-Nearest Neighbors, Support Vector Machine, and XGBoost. Models are tested with accuracy metric, precision metric, and F1 score. Then, results of evaluation are studied to see their performance at identifying the best classification model, which also gives insight into how effective the classification might be. The concluding outcomes are seen in the results section to understand collectively the classification framework. Different classification algorithms, used are- random forest, k-nearest neighbors, support vector machine and XGBoost. [21] has reviewed various ML algorithms for various applications and highlighting the advantages and limitations of those models. Some of the models have been taken into consideration for the review.

3.1 Random forest

As the name implies, a random forest is made up of a huge number of individual decision trees that work together as an ensemble. It is based on the combination of two key principles:

1. Only a subset of training data is used to train each decision tree.
2. Only a random subset of the features is allowed to be used in each decision tree's prediction.

Each tree in the random forest produces a class prediction, and the class with the most votes becomes the prediction of our models.

3.2 K-nearest neighbour

K-Nearest Neighbour is a simple Machine Learning method that uses the Supervised Learning methodology to handle classification and regression issues. It's a lazy learner algorithm that's non-parametric. KNN stands for the number of closest neighbours. The quantity of neighbours is the most important consideration. KNN works by locating the k nearest points to the goal and classifying them based on the majority votes of their k neighbours. Each item votes for their class, and the prediction is determined by the class with the most votes. Distance between points is determined using distance metrics such as Euclidean distance to identify the closest similar points. KNN has been used in various

applications of classification like intent creation and detection [22].

3.3 Support vector machine

Support Vector Machine is a supervised machine learning technique that is mostly used to solve classification issues. Each data element is displayed as a point in n-dimensional space (where n is the number of properties). This means that the value is the value at a specific coordinate of the SVM algorithm. Next, vertical classification by finds hyperlevels and clearly distinguishes between the two classes. SVMs are used on binary classification in their most basic form, separating data points into 1 or 0. The same principle is applied to multiclass classification. Multiple binary classification cases, often known as one-vs-one, are used to break down the multiclass problem. The goal of the SVM method is to maximize the distance between the data and Hyperbelle. Hinge loss is a useful loss function associated with maximizing margins. Control parameters are also added to the cost function. The parameters are to maximize margins and achieve loss compromises. The cost function looks like this after adding the regularization argument.

3.4 XGBOOST

Another ensemble classifier is boosting, which is created by combining a number of weak learners. Forward stagewise additive modelling is used to train boosting algorithms in a sequential fashion. XGB stands for extreme Gradient Boosting, which is a type of boosting that allows you to train your model quickly and in parallel. A gradient boost is a technology in which a new model is created to predict residuals or errors from previous models, then combined into the final model. forecast. It's termed gradient boosting because it employs a gradient descent technique to reduce the size of the data set.

Complex models can be penalized using both L1 and L2 regularization in XGBoost. Overfitting can be avoided by regularization.

4. EXPERIMENTS AND RESULTS

This section discusses about analysis of various ML algorithms discussed in previous section.

4.1 Music genre classification

In a random forest model, there are three parameters for which a few different values were used and chose the ones for which the accuracy came out to be best n estimators specify the number of trees wanted to build before making predictions and max_depth specifies the depth of each tree (should be in range 1 to 32). The larger the number of trees and deeper the height, the better is the accuracy. After many trials, the final value is chosen for parameters as n estimators = 1000, max depth = 10 and random state = 10. In k-nearest neighbours, the parameter used is the n neighbours i.e. the value of k. A small value of k means that noise will have a higher influence on the result and a large value makes it computationally expensive. So, the value must be odd and for better results, $k = \sqrt{n}$, formula can

be used where n is the number of samples in the training set. Since our training set contains 80 images, and took the value of k as 9. In the Support vector machine, there is only one parameter specified i.e. decision function shape. One-vs-one for this parameter of multiclass is chosen as problem. In XGBoost, there are two parameters specified, n estimators and learning rate. The learning rate is the tuning parameter of the optimization algorithm and defines the step size of all iterations if the algorithm moves to the minimum loss function in the direction of the loss function. The value of the learning rate should be small (around 0.1 - 0.3), so the value as 0.3 is choose and n estimators = 1000. Cosine similarity is a metric used to measure how is two similar elements. Mathematically, it measures the cosine of the angle between two vectors projected into multidimensional space. The starting value range is 0 1. A function is also defined to capture 5 most similar values to the input audio file and build the recommendation system.

4.3 Metrics

Based on some metrics the results are evaluated to know which model performs well. The following are the metrics:

Accuracy - It's only the proportion of accurately anticipated observations to total observations.

Precision - The ratio of accurately predicted positive observations to total expected positive observations is known as precision.

F1 Score - The weighted average of Precision and Recall is the F1 Score.

Recall - The ratio of accurately predicted positive observations to all observations in the actual class is known as recall.

4.4 Performance of models

• Random Forest

Table 1: Results obtained by Random Forest model

Genre	precision	recall	F1 score
Blues	0.85	0.83	0.84
Classical	0.92	0.96	0.94
Country	0.70	0.71	0.71
Disco	0.74	0.79	0.76
HipHop	0.86	0.78	0.82
Jazz	0.83	0.88	0.85
Metal	0.86	0.94	0.90
Pop	0.90	0.79	0.84
Reggae	0.69	0.85	0.76
Rock	0.77	0.60	0.67
Accuracy			0.81

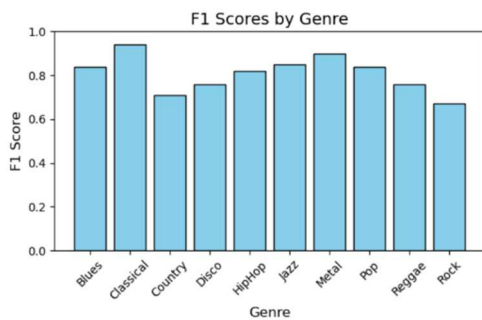


Fig 3: Graphical visualization of Random Forest model for F1-Score

• K-Nearest Neighbors

The F1 score on various Genre using KNN can be shown in Table 2 with its graphical visualization in Figure 4.

Table 2: Results of K- Nearest Neighbors model.

Genre	precision	recall	F1 score
Blues	0.93	0.93	0.93
Classical	0.92	0.94	0.93
Country	0.85	0.85	0.85
Disco	0.83	0.91	0.87
HipHop	0.92	0.85	0.89
Jazz	0.86	0.87	0.86
Metal	0.97	0.93	0.95
Pop	0.93	0.86	0.89
Reggae	0.80	0.90	0.85
Rock	0.83	0.81	0.82
Accuracy			0.89

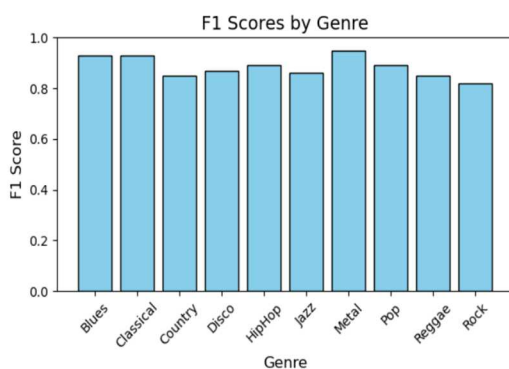


Fig 4: Graphical visualization of KNN model for F1-Score

• Support Vector Machine

The F1 score on various Genre using SVM can be shown in Table 3 with its graphical visualization in Figure 5.

Table 3: Results obtained by Support Vector Machine model.

Genre	precision	recall	F1 score
Blues	0.72	0.80	0.76
Classical	0.89	0.92	0.90
Country	0.67	0.67	0.67
Disco	0.66	0.71	0.68
HipHop	0.79	0.68	0.73
Jazz	0.78	0.81	0.79
Metal	0.82	0.88	0.85
Pop	0.86	0.82	0.84
Reggae	0.67	0.74	0.70
Rock	0.59	0.45	0.51
Accuracy			0.75

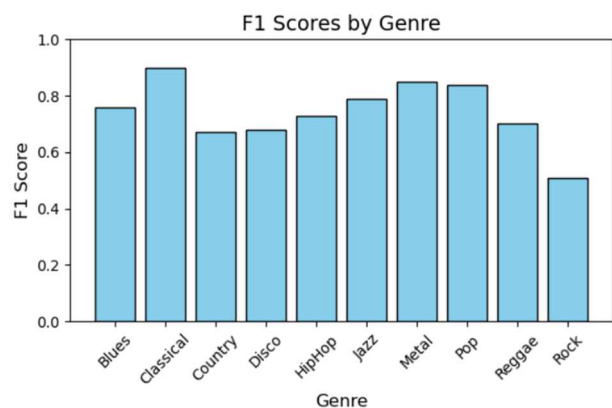


Fig 5: Graphical visualization of SVM model for F1-Score

• XGBoost

The F1 score on various Genre using XGBoost can be shown in Table 4 with its graphical visualization in Figure 6.

Table 4: Results obtained by XGBoost model

Genre	precision	recall	F1 score
Blues	0.92	0.94	0.93
Classical	0.94	0.96	0.95
Country	0.85	0.91	0.88
Disco	0.91	0.92	0.91
HipHop	0.94	0.93	0.94
Jazz	0.91	0.90	0.90
Metal	0.97	0.97	0.97
Pop	0.95	0.94	0.94
Reggae	0.91	0.93	0.92
Rock	0.90	0.79	0.84
Accuracy			0.92

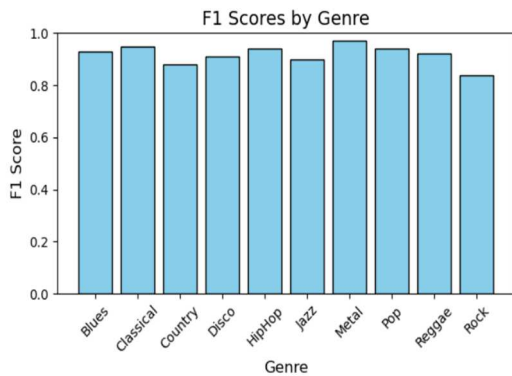


Fig 6: Graphical visualization of KNN model for F1-Score

5. CONCLUSION

A robust model was established to classify audio files into 10 different genres and recommend top songs that relate to the input audio. Genre classification was approached using two primary methods: traditional algorithms for classification and deep neural networks. Among all the classification algorithms, random forest, k-nearest neighbors, support vector machine, and XGBoost, the latter had been on top at 92% accuracy. This was significantly contributed by the advanced boosting technique in XGBoost that contains regularization in the decrease in overfitting and thus generalization. A fine model like this speaks of success in the tuning and optimization in the boosting algorithms, thus making XGBoost most reliable for this task.

The results indicate the successes in machine learning area of genre classification for music and song recommendation, as well as indicating areas that require increased advancements with additional data and better model architectures.

6. REFERENCES

- [1] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, Jul. 2002, doi: 10.1109/TSA.2002.800560.
- [2] Z. Cataltepe, Y. Yaslan, and A. Sonmez, "Music genre classification using MIDI and audio features," *EURASIP J Adv Signal Process*, vol. 2007, 2007, doi: 10.1155/2007/36409.
- [3] H. Bahuleyan, "Music Genre Classification using Machine Learning Techniques," Apr. 2018, [Online]. Available: <http://arxiv.org/abs/1804.01149>
- [4] A. Meng, P. Ahrendt, J. Larsen, and L. K. Hansen, "Temporal feature integration for music genre classification," *IEEE Trans Audio Speech Lang Process*, vol. 15, no. 5, pp. 1654–1664, Jul. 2007, doi: 10.1109/TASL.2007.899293.
- [5] "IJRESM_V4_I7_17".
- [6] "MUSIC GENRE CLASSIFICATION USING MACHINE LEARNING," 2021. [Online]. Available: www.sathyabama.ac.in
- [7] "2022_Vol13_No1_it3a".
- [8] "MUSIC GENRE CLASSIFICATION AND RECOMMENDATION SYSTEM," *International Research Journal of Modernization in Engineering Technology and Science*, May 2023, doi: 10.56726/irjmets40242.
- [9] A. Elbir, H. Bilal Çam, M. Emre Iyican, B. Öztürk, and N. Aydin, "Music Genre Classification and Recommendation by Using Machine Learning Techniques," in *Proceedings - 2018 Innovations in Intelligent Systems and Applications Conference, ASYU 2018*, Institute of Electrical and Electronics Engineers Inc., Nov. 2018. doi: 10.1109/ASYU.2018.8554016.
- [10] J. Reed and C.-H. Lee, "A Study on Music Genre Classification Based on Universal Acoustic Models," 2006.
- [11] E. Benetos, Q. Mary, and C. Kotropoulos, "A tensor-based approach for automatic music genre classification," 2008. [Online]. Available: <https://www.researchgate.net/publication/50342245>
- [12] L. Breiman, "Random Forests," 2001.
- [13] S. Chillara, K. A. S, S. A. Neginhal, S. Haldia, and V. K. S, "Music Genre Classification using Machine Learning Algorithms: A comparison," *International Research Journal of Engineering and Technology*, 2008, [Online]. Available: www.irjet.net
- [14] J. Dias, H. Deshmukh, V. Pillai, and A. Shah, "Music Genre Classification & Recommendation System using CNN." [Online]. Available: <https://ssrn.com/abstract=4111849>
- [15] J. Mendes, "Deep Learning Techniques for Music Genre Classification and Building a Music Recommendation System MSc Research Project Data Analytics."
- [16] *Proceedings: 2014 IEEE International Conference on Systems, Man and Cybernetics (SMC) : October 5-8, 2014 : San Diego, CA, USA*. [Institute of Electrical and Electronics Engineers], 2014.
- [17] G. R. Arce, Y. Panagakis, and C. Kotropoulos, "Music genre classification via sparse representations of auditory temporal modulations," 2009. [Online]. Available: <https://www.researchgate.net/publication/50342516>
- [18] "18".
- [19] F. Wang, S. Fu, and F. Abza, "Rigdelet neural network and improved partial reinforcement effect optimizer for music genre classification from sound spectrum images," *Heliyon*, vol. 10, no. 14, Jul. 2024, doi: 10.1016/j.heliyon.2024.e34067.
- [20] İ. Tarimer and B. C. Karadağ, "Genres Classification of Popular Songs Listening by Using Keras," *Gazi University Journal of Science Part A: Engineering and Innovation*, vol. 11, no. 1, pp. 123–136, Mar. 2024, doi: 10.54287/gujisa.1374878.
- [21] S. Gupta, M. Kaur, S. Lakra, Y. Dixit, "A Comparative Theoretical and Empirical Analysis of Machine Learning Algorithms," in *Webology*. 2020 Jun;17(1):377-97.
- [22] C. Magoo, M. Singh, "An implemented review for intent creation using different clustering techniques," in *2022 Fifth International Conference on Computational Intelligence and Communication*

Technologies (CCICT) 2022 Jul 8 (pp. 83-91).
IEEE.